

Lab: Derivation Exercise — Free Energy and Markov Blankets

Objective

Derive the core mathematical results underlying Active Inference: the Markov Blanket partition, the variational free energy bound, and the decomposition of free energy into accuracy and complexity.

Part 1: Markov Blanket Derivation

Goal: Prove the conditional independence conditions from the graphical model.

Given a Bayesian network with four variable sets: internal (μ), external (η), sensory (s), active (a), where:

- $\eta \rightarrow s$ (external states cause sensory states)
- $\mu \rightarrow a$ (internal states cause active states)
- $s \rightarrow \mu$ (sensory states influence internal states)
- $a \rightarrow s$ (active states influence sensory states)

Derivation task: Using the d-separation criterion, prove that $\mu \perp \eta \mid \{s, a\}$.

Show all steps, starting from the graphical structure and applying the d-separation rules (blocking via conditioning on colliders vs. non-colliders).

Write your response here

Part 2: The Free Energy Bound

Goal: Derive the ELBO inequality.

Starting from:

$$F = E_q[\ln q(\eta) - \ln p(\eta, s)]$$

Step 1: Expand $p(\eta, s) = p(\eta|s) \cdot p(s)$

Step 2: Substitute and rearrange to obtain:

$$F = D_{KL}[q(\eta) \parallel p(\eta|s)] - \ln p(s)$$

Step 3: Since $D_{KL} \geq 0$, conclude $F \geq -\ln p(s)$

Write out each step of the derivation completely, showing all algebraic manipulations.

Write your response here

Part 3: Accuracy-Complexity Decomposition

Goal: Derive the alternative decomposition of free energy.

Starting from $F = E_q[\ln q(\eta) - \ln p(\eta, s)]$, show that:

$$F = -E_q[\ln p(s|\eta)] + D_{KL}[q(\eta) \parallel p(\eta)]$$

where:

- $-E_q[\ln p(s|\eta)]$ is the *negative accuracy* (expected negative log-likelihood)
- $D_{KL}[q(\eta) \parallel p(\eta)]$ is the *complexity* (divergence of posterior from prior)

Explain why minimizing F requires balancing accuracy (explaining the data) with complexity (not deviating too far from priors).

Write your response here

Part 4: Gaussian Example

Goal: Compute free energy analytically for Gaussian distributions.

Let the generative model be:

$$p(\eta) = N(\eta; \mu_0, \sigma_0^2) \quad p(s|\eta) = N(s; \eta, \sigma_s^2)$$

And the recognition density be: $q(\eta) = N(\eta; \mu_q, \sigma_q^2)$

Task: Compute $F = -E_q[\ln p(s|\eta)] + D_{\text{KL}}[q(\eta) \parallel p(\eta)]$ analytically. Show all integrals.

Then find the optimal μ_q and σ_q by setting $\partial F / \partial \mu_q = 0$ and $\partial F / \partial \sigma_q = 0$.

Write your response here

Part 5: Synthesis

Goal: Connect the mathematical results to the conceptual framework.

In 200 words, explain how the three derivations (Markov Blanket, ELBO, accuracy-complexity) together establish the mathematical foundation of Active Inference: that a self-organizing system with a Markov Blanket can always be described as minimizing free energy, which entails approximate Bayesian inference.

Write your response here

Lab Summary

Part	Skill Developed	Mathematical Result
1	Graphical model reasoning	Markov Blanket conditional independence proof
2	Variational calculus	ELBO derivation and bound proof
3	Decomposition	Accuracy-complexity trade-off
4	Analytic computation	Gaussian free energy optimization
5	Mathematical-conceptual integration	Connecting derivations to Active Inference